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A Quick End to a Long Story: Networks and Mexican Migration during the Great Recession

By
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During times of economic recession, migrants face long periods of unemployment or underemployment in destination countries. This information is transmitted to migrant-sending households via networks that link communities of origin and destination, letting potential migrants know that if they were to migrate they would likely experience low and unstable earnings, and that remittances normally expected from international migration might be placed at risk. In this event, there are few incentives for other household members to migrate. This study examines the effect that information sent through networks during recessionary times has in reducing the likelihood of out-migration, thereby explaining why Mexico-U.S. migration fell so suddenly with the onset of the Great Recession in the United States.

Keywords: networks; migration; information; recession; risk

The theory of migrant networks developed by Taylor (1986) in his article "Differential Migration, Networks, Information and Risk" argues that household income in rural Mexico consists of money derived from four basic sources: earnings contributed by household members working in family enterprises, earnings contributed by members working in local jobs, earnings contributed by members working elsewhere in Mexico; and earnings contributed by household members working abroad, which, in the case of Mexico, is overwhelmingly the United States. The economic problem for the household is to decide how best to allocate household workers across these four productive activities.

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At the time the decision-maker (household-head) formulates the household labor assignment strategy, he does not know with certainty what the earnings from each potential productive activity might be. Efficiency in assigning household members to labor force options depends on the quality of information available to the decision-maker about the amount of money to be earned from each activity, the variability of the resulting earnings, and the risks associated with those earnings (Taylor 1986, 148). Whichever option maximizes expected income while minimizing income variability and income risk will perforce maximize the utility the household expects to derive from that option and thereby incentivize the household-head to invest the household's labor resources in that activity (Taylor 1986, 157).[1]

Given that migrant networks offer information about potential earnings available at destinations where household and community members migrate, access to these networks reduces the *risk* to income earned at those places. Migrant networks also offer information to help migrants find better-paying jobs at places of destination, raising the *amount* of the expected income to be gained through migration; and, since networks offer a structure of social support to help migrants locate jobs with more reliable earnings, they also reduce the *variability* of income from migration (Taylor 1986, 149). These considerations render the expected utility derived from migration relatively attractive compared with other productive options, thus increasing the likelihood that the household will send one or more members out to work as migrants, and, in doing so, reduce the risks to income while maximizing the potential gains.

The research presented here applies Taylor's social network theory to demonstrate the influence that information from migrant networks has on the labor allocations of households during times of recession. After introducing the problem, I apply his model to consider what happens when migrant networks connect households to U.S. destinations experiencing an economic downturn. I focus on the labor market circumstances that prevailed in the United States after the onset of Great Recession in late 2007 and document the downward shift in the volume of migration between Mexico and the United States that occurred after that date. Using data from the Mexican Migration Project (MMP), I then undertake an empirical analysis demonstrating the influence of networks on migratory decisions taken before and after the onset of the Great Recession. I close by offering a summary of conclusions and a discussion of their implications.

Network Effects during Recessionary Times

Taylor's (1986) approach holds that household income (Y) comprises the sum total of the earnings contributed by members engaged in four different kinds of productive activities: work in a family business enterprise; work at a paying job in the local community; work at a job located somewhere else in the same country; and work abroad in another country. Each household-head has to decide how to distribute its working members across these four activities, but the returns

expected from each activity are not known with certainty at the time that the decision must be made.

For example, earnings derived from labor on a family farm are influenced by variables outside the household's control, such as droughts, floods, and plagues, which affect crop productivity, as well as by changes in the price of farm inputs and outputs related to shifting market conditions. The income derived from wages earned from jobs in the community of origin likewise may vary with shifts in the local labor market; and, finally, the earnings expected from migrants working elsewhere depends on conditions in national and international markets, as well as on the ability of migrants to travel to different destinations, find a job, remain employed, and evade detention and deportation by immigration authorities if they are unauthorized. In addition, the amounts received by the sending household depend on the propensity of migrants to remit and save; and all of these eventualities are of course subject to stochastic shocks (Taylor 1986, 154).

Here, I seek to model the effect of migrant networks on the decision to migrate during times of economic recession in countries of destination. Given multiple sources of uncertainty, household-heads cannot maximize expected earnings in the way normally assumed by traditional neoclassic economic models of migration (such as that developed by Todaro [1969]). Instead, households must take into consideration not only the income expected from different productive activities, but also the stability and risks associated with each source of income. They must then select the economic activity that offers the greatest gain at the smallest risk with the least variability, just as investors seek to balance gains, risks, and variability in their stock portfolios (Taylor 1986, 153).

Taylor assumes that households are risk averse, such that $u'(\cdot)>0$ and $u''(\cdot)<0$. He also assumes that the structure of household preferences complies with a Neumann-Morgenstern utility function, whose only argument is total household income $u(\mathrm{Y})$ (Von Neumann and Morgenstern 1944, 26). Given that Y is randomly distributed, maximizing utility is achieved by maximizing expected income:

$$\mathrm{E}[u(Y)] = \sum_{i=1}^{n} \mathrm{p}(Y_i)u(Y_i) \tag{1}$$

The effect of risk aversion is incorporated into the expected utility function by replacing $u(Y)$ with a second-order Taylor expansion series[2] around $E[(\mathrm{Y})]=\hat{\mathrm{Y}}$, and taking the expected value of the resulting expression:

$$E\left[u(\mathrm{Y})\right] = u\left(\hat{\mathrm{Y}}\right) + E\left[\mathrm{Y} - \hat{\mathrm{Y}}\right]u'\left(\hat{\mathrm{Y}}\right) + ½E\left[\left(\mathrm{Y} - \hat{\mathrm{Y}}\right)^2\right]u''\left(\hat{\mathrm{Y}}\right) + \ldots, \tag{2}$$

in which a random distribution of Y implies that:

$$E\left[\left(\mathrm{Y} - \hat{\mathrm{Y}}\right)\right] = 0 \text{ and } E\left[\left(\mathrm{Y} - \hat{\mathrm{Y}}\right)^2\right] = \sigma_{\mathrm{Y}}^2. \tag{3}$$

If the variance of income (σ_Y^2) is relatively small, then we can disregard the higher order terms to arrive at the following approximation:[3]

$$E\left[u(Y)\right] = u\left(\hat{Y}\right) + \tfrac{1}{2}\sigma_Y^2 u''\left(\hat{Y}\right). \tag{4}$$

In this way, expected utility can be decomposed into two components, the first of which is the utility of income $u(\hat{Y})$ and the second of which is the degree of risk $\tfrac{1}{2}\sigma_Y^2 u''(\hat{Y})$, which equals the variability in household income σ_Y^2 weighted by the degree of risk $u''(\hat{Y})$ (Taylor 1986, 153; David 1974, 33).

Given that income is desirable, the first component is positive and the second is negative, assuming that the household is risk averse (i.e. $u''(\cdot)<0$). As a result, any changes in explanatory variables that raise the expected income and reduce the variability and risk of a specific economic activity improves its expected utility relative to other activities so that the household-head is prompted to invest its labor effort in that option (Taylor 1986, 157).

The problem then becomes one of choosing a labor option d^{*j} for individual j that maximizes the future flow of expected utility for the household, discounted by the function $z(\cdot)$, a labor allocation problem summarized by the following decision rule:[4]

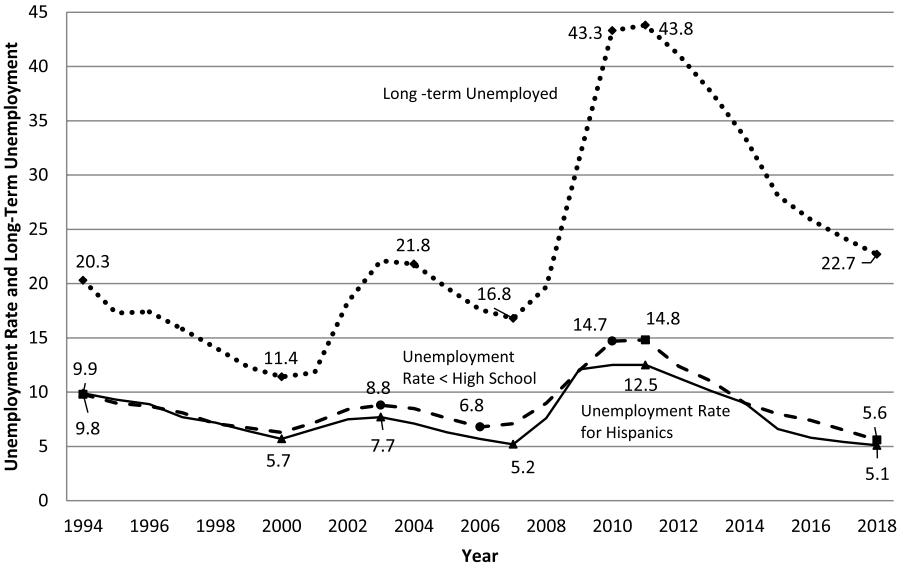
$$d^{*j} = \begin{cases} 1 & \text{if } z_{d^*}^j > z_d^j \text{ for all } d \neq d^*, \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where $d = 1,\ldots d$ is defined as the set of distinct labor options available to the household. In this equation, $z_{d^*}^j$ is defined as the activity d^* allocated to household member j that maximizes the future flow of expected utility for the household, taking into account both individual and household characteristics. Likewise, z_d^j is defined as the future flow of expected utility for the household if member j is assigned to work in activity d (Taylor 1986, 153; 1984a, 36; 1984b, 10).

Since migrant networks offer information to households about employment opportunities and wages at specific places of destination, they reduce the risk associated with the income expected from sending one or more members to those places. At the same time, migrant networks offer migrants information that helps them to find better-paying jobs, raising the income expected from migration. Finally, the variability of the earnings obtained through migration is reduced thanks to the infrastructure of social support that migrants can count on at places of destination.

As a result, sending a household member to a destination well-connected to the household by network ties increases its expected income and reduces its variability and risk, thereby raising the expected utility of that activity relative to other productive options and incentivizing the household-head to invest its labor effort in migration. The foregoing explains why migrant networks increase the probability that household members will go to work in destinations where they

FIGURE 1
Trends in the Unemployment Rate for Hispanics and Workers with less than High School Degree and Percent Unemployed 27 Weeks or Longer, 1995–2018



SOURCE: U.S. Bureau of Labor Statistics (2018). See https://data.bls.gov.

can count on network-based social capital, while simultaneously reducing the likelihood that they stay home or migrate to other localities that lack this kind of social capital (Taylor 1986, 157).

In sum, Taylor's theoretical approach establishes that migratory decisions depend on income expected from migration relative to other employment options, the variability of that expected income, and the risks associated with migration and employment at the place of destination (along with the costs of migration). Households invest their labor resources in migration when it offers the highest income, lowest variability, and least risk. In contrast, households avoid investing in migration when that option implies lower earnings with greater variability and higher risks. This last aspect of the determinants of migration is what connects the decision to migrate to macroeconomic circumstances in the country of destination.

During times of economic recession at places of destination, newly arriving migrants will face unemployment, underemployment, or limited hours of work, which implies a reduction in expected income, greater variability in income over time, and higher risks to income. When this information is transmitted to households in Mexico via networks, households know that the income from migration will likely produce lower returns, greater variability, and higher risk, making it more attractive to invest labor in less risky options elsewhere that offer greater and more reliable access to income. The transmission of information about wages and employment conditions in the United States back to households in sending

TABLE 1
Percent Latino and Percent Unemployed, by Industrial Sector, 2014

Sector	Percent Latinos	Percent Unemployed
Construction	27.3	8.3
Agriculture, Forestry, etc.	23.1	12.1
Leisure and Hospitality	22.3	7.5
Other Services	19.0	5.4
Extractive Industries	18.6	4.1
Transportation	17.2	4.2
Commerce	16.4	5.3
Professional Services	16.0	5.8
Manufacturing	15.8	3.9
Education and Health	11.5	3.8
Public Administration	11.4	2.8
Financial Activities	11.3	3.4
Information Technology	10.5	5.7

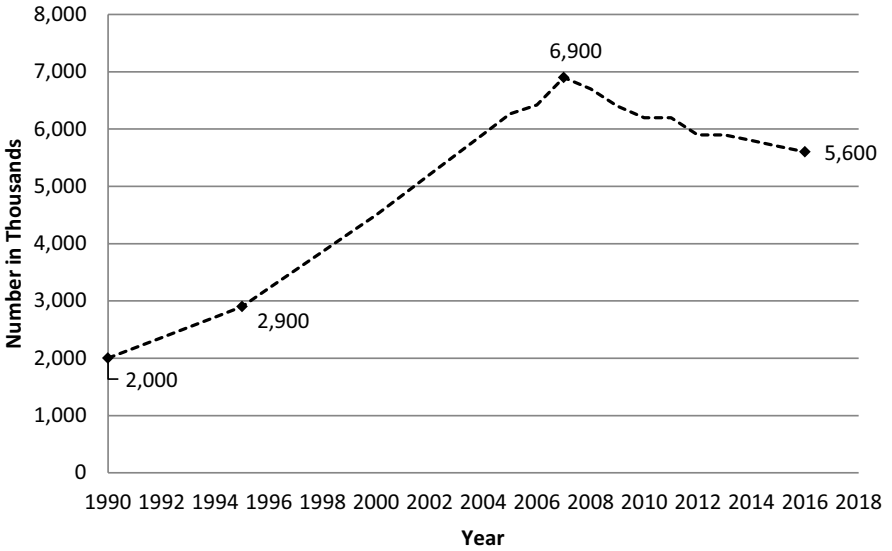
Source U.S. Bureau of Labor Statistics. https://www.bls.gov/

communities explains, at least in part, why undocumented Mexico-U.S. migration ceased abruptly with the onset of the Great Recession.

Analysis of the U.S. Labor Market

The Great Recession began in late 2007, and depressed labor market circumstances persisted over the next seven years. The solid line in Figure 1 shows the effect of the recession on the unemployment rate observed for Latinos in the United States. Beginning from a low point of 5.2 percent in 2007, the unemployment rate rose to peak at 12.5 percent in 2011 and did not fall below 10 percent until 2014. Thereafter, unemployment steadily dropped until it reached 5.1 percent in 2018, just below its pre-recession minimum of 5.2 in 2007. The large majority of Mexican immigrant workers in the United States have less than a high school education, so Figure 1 also plots the trend in unemployment rates for workers at this education level. For these less-educated workers, the pre-recession minimum rate of 6.8 percent was reached in 2006, a year before that of Latinos as a whole. Thereafter, the unemployment rate rose faster and peaked earlier at 14.8 percent in 2011, more than two points above the overall rate for Latinos. Not only did rates of unemployment rise, but the duration of time spent without employment rose even more sharply. As shown by the dotted line in Figure 1, the long-term unemployment experienced by U.S. workers rose from a minimum of 16.8 percent in 2007 to peak at 43.8 percent in 2011 before declining to 22.7 percent in 2018.

FIGURE 2
Estimated Size of Mexican Undocumented Populations of the United States



SOURCE: Pew Research Center (2017). See https://www.pewresearch.org.

Mexican immigrant workers tend to work in industrial sectors that were hit particularly hard by the recession. Although we cannot focus specifically on Mexican immigrants, Table 1 ranks industrial sectors by the percentage of Latino workers within them in 2014, revealing exceptionally high shares in construction (27.3 percent), agriculture (23.1 percent), and leisure and hospitality (22.3 percent). These sectors were followed closely by the category "other services" (19 percent), extractive industries (18.6 percent), and transportation (17.2 percent). The second column shows the unemployment rate recorded in each of these categories in 2014, and it is immediately clear that those sectors of the economy where Latinos were overrepresented were also those characterized by the highest rates of unemployment in the wake of the Great Recession. In that year, the top four categories for Hispanic workers displayed unemployment rates of 8.3, 12.1, 7.5, and 5.4 percent, and the simple correlation between percent Hispanic and the unemployment rate across industrial sectors was 0.73.

Thus, the onset of the Great Recession in the United States caused unemployment to rise sharply in sectors that employed Latinos and less educated workers, bringing about a sharp increase in the duration of unemployment for all workers that persisted through 2014. Being among the most vulnerable to the effects of an economic downturn, unauthorized Mexican migrants in the United States would surely be aware of these shifts in economic circumstances and likely would have transmitted this information to friends and family members in communities of origin. This information would then quickly have spread through

social networks within the communities to update local knowledge of community residents about the income ($u(\hat{Y})$), income variability ($\frac{1}{2}\sigma_Y^2$), and risks to income ($u''(\hat{Y})$) that had come to be associated with labor migration to the United States.

Although information about the new economic realities likely traveled southward rapidly via migrant networks, many U.S. migrants also lost their jobs, and, with no means of support, they returned home to bear witness in person about the changed economic circumstances north of the border. As shown in Figure 2, after steady growth from 1990 onward the undocumented Mexican population of the United States peaked at 6.9 million in 2007 and abruptly reversed course to proceed steadily downward in the years thereafter. According to the estimates of Passel and Cohn (2017), the number of undocumented Mexican residents fell from 6.9 million in 2007 to 5.6 million in 2017, indicating the net out-migration of 1.3 million former undocumented migrants.

Methods and Data

In 1987, Massey and García-España specified and estimated a logistic regression model to predict the likelihood of migration to the United States as a function of ties to the migrant network, controlling for other individual, household, and community characteristics. Here, I adapt their model to demonstrate empirically the effect of information transmitted to households via migrant networks on migration to the United States by estimating the following equation:

$$P_i\left(y_i = 1\right) = exp(\beta x_i)/\left[1 + exp\left(\beta x_i\right)\right], \tag{6}$$

where P_i is the probability of migrating to the United States, x is a vector of explanatory variables, and B is a vector of regression coefficients associated with each of these variables.

To estimate the foregoing equation, I use data compiled by the MMP on first trips to the United States and select households in communities 72 through 161, which were surveyed in the years just before and after the onset of the Great Recession in 2007. From the resulting set of households, I further select those in which at least one person listed on the household roster was present in the United States during 2000–2016. I then follow households forward in time from the date of the first member's departure to the point at which another member left for the United States or the survey date, whichever came first. Measuring time using six-month intervals, I code the dependent variable as 1 if another household member departed during the interval, and 0 otherwise, excluding all intervals after this second departure.

I then used a logistic regression model to predict the departure of an additional household member from a set of independent variables (the x-vector in the above equation) defined in Table 2. The key predictor is the unemployment rate prevailing for Hispanics in the United States during the time interval. Given that each household already had a member residing in the United States, information

TABLE 2
Variables Used in Analysis of Migration during Economic Recessions

Variables	Definition
U.S. Economic Context	
Unemployment Rate	Rate of Hispanic Unemployment in the United States
Trip Duration in Years	
First	1 if 1–2 semesters, 0 otherwise
Second	1 if 3–4 semesters, 0 otherwise
Third	1 if 5–6 semesters, 0 otherwise
Fourth	1 if 7–8 semesters, 0 otherwise
Fifth and Greater	1 if 9 semesters or greater, 0 otherwise
Demographic Background	
Age	Age at last birthday
Age^2	Age squared
Sex	1 if male, 0 otherwise
Rural Origins	1 if origin community less than 2500, 0 otherwise
Human Capital	
Education	Years of schooling
Sector of Employment	
Construction Worker	1 if worked in US construction industry, 0 otherwise
Legal Status	
Undocumented	1 if lacks legal right to work in U.S., 0 otherwise
Household Property	
House Owner	1 if the dwelling is owned, 0 otherwise
Land or Business Owner	1 if household owns cultivable land or business, 0 otherwise
Household Composition	
Number of workers	Number of workers in household
Community Investment Climate	
Numbers of Banks	Number of bank branches in municipality
Cost of Migration	
Smuggling Costs	Cost of border crossing in 2017 U.S. dollars
Location of Interview	
U.S. Resident	1 if interviewed in U.S., 0 otherwise

about U.S. labor market conditions is expected to be transmitted readily back to family members in Mexico. During periods of economic recession, this information is predicted to reduce the likelihood that another household member would depart for the United States.

The effect of the Hispanic unemployment rate on the likelihood of additional migration is assessed while controlling for a variety of individual, household, and community characteristics that have been used in prior estimation models (Massey and García-España 1987; Lindstrom 1996; Massey and Espinosa 1997).

Because it is an event history analysis, the model includes dummy variables indicating the duration of time since the first household member left for the United States. The model also controls for the first household migrant's demographic characteristics (age, gender, rural origins) and human capital attainments (years of education). With respect to the departed migrant, the model holds constant the effect of legal status (undocumented or not) and the sector of U.S. employment (whether in construction, the hardest hit of all sectors during the Great Recession; see Rugh 2015). Additional controls include indicators of household composition (number of workers), community investment climate (number of branch banks), the costs of migration (smuggling costs in 2017 U.S. dollars), and the location of the interview (in the United States or not).

Networks and Migration during Recessionary Times

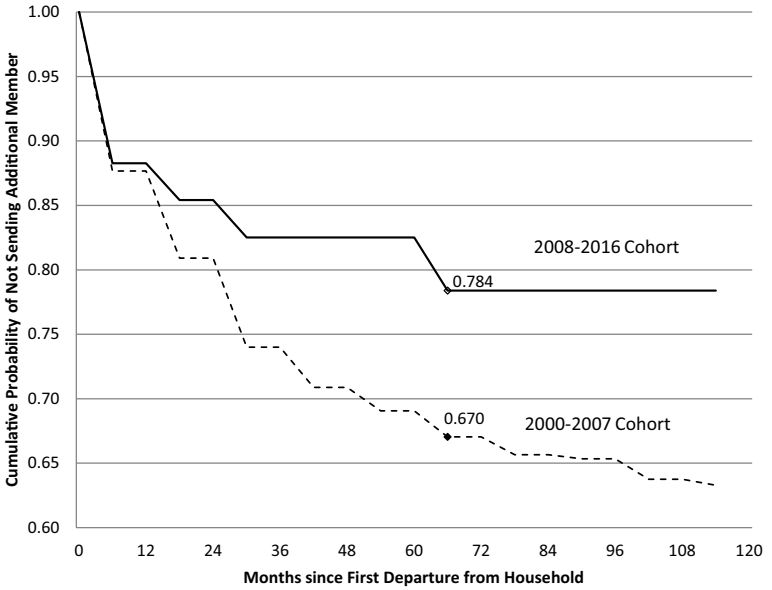
Table 3 contains two sets of columns. To the left are means and standard deviations for all independent variables, and to the right are coefficients indicating estimated effects of these variables on the likelihood of sending an additional household member to the United States. The typical departed migrant was aged 28 years and male, with 8.4 years of schooling. Around 5 percent were interviewed in the United States and 30 percent were of rural origin, originating with communities containing an average of 3.5 bank branches. The large majority of households, 85 percent, lived in owned homes and 22 percent reported owning cultivable land or a business enterprise. Around 19 percent of migrants present in the United States worked in construction and 87 percent were undocumented, paying an average of U.S. $2,920 to gain entry to the United States. During the average time interval (2000–2016), the observed Hispanic unemployment rate was 7.5 percent.

The columns on the right present coefficients and standard errors estimated to capture the effect of each variable on the likelihood of sending an additional migrant, given that one household member was already present in the United States. Consistent with expectations, the coefficient estimate for the Hispanic unemployment rate was strongly and significantly negative. When households have a member working in the United States and unemployment there rises, the likelihood of sending an additional member northward is decisively reduced. With each point increase in the unemployment rate, the likelihood of an additional departure falls by around 11 percent. Beyond this effect, the likelihood of an additional departure also falls over time, and, as crossing costs increase, the likelihood varies in curvilinear fashion with the age of the departed migrant, and is systematically lower when the first migrant is male and works in construction. Additional out-migration is more likely among households that contain more workers, own their dwellings, and possess land or a business enterprise, consistent with the view that migration is used strategically as a means of diversifying sources of income and to manage risks to household well-being (Stark 1978; Taylor 1986).

TABLE 3
Descriptive Statistics and Estimates of Equation Predicting the Sending of an Additional Household Member to the United States

	Descriptive Statistics		Household Sends an Additional Migrant to the U.S.	
Independent Variables	Mean	Standard Deviation	B	SE
U.S. Economic Context				
Hispanic unemployment rate	7.478	2.087	–0.112***	0.039
Trip Duration in Years				
First	0.288	0.453	---	---
Second	0.192	0.394	–0.529***	0.141
Third	0.129	0.336	–0.405***	0.160
Fourth	0.091	0.288	–1.154***	0.248
Fifth and Greater	0.299	0.458	–1.709***	0.228
Demographic Background				
Age	28.300	8.910	0.077***	0.035
Age^2	880.261	638.732	–0.001***	0.001
Male	0.738	0.440	–0.377***	0.121
Rural Origins	0.303	0.459	–0.035	0.117
Human Capital				
Education	8.423	3.184	–0.026	0.017
Sector of Employment				
Construction Worker	0.185	0.388	–0.454***	0.166
Legal Status				
Undocumented	0.868	0.338	–0.262	0.159
Household Property				
Home Owned	0.847	0.360	0.414***	0.172
Land or Business Owned	0.218	0.413	0.278***	0.120
Household Composition				
Number of workers	1.780	1.206	0.126***	0.040
Community Investment Climate				
Numbers of Banks	3.462	14.713	0.0005	0.003
Cost of Migration				
Smuggling Costs	2920	854	–0.0003***	0.000
Location of Interview				
U.S. Resident	0.046	0.210	0.653***	0.254
Constant			–2.402***	0.668
Likelihood Ratio			–1506.2307	
LR Chi Squared			248.210	
Pseudo R-Squared			0.0761	
N			10,664	

FIGURE 3
Survival Curves for Sending an Additional Household Member to the United States for Mexican Households in Two Cohorts



Conclusion

Until now, the theory of migrant networks has not been applied to analyze household decision-making during times of economic recession in countries of destination, when rates of unemployment rise, hours of work fall, and durations in unemployment increase. Under these circumstances, migratory networks that connect migrants in the United States to people from communities of origin readily transmit this information, so that households quickly learn that earnings expected from migration are likely to be lower, riskier, and less reliable, incentivizing households to allocate household workers to other productive activities, either in the locality or elsewhere in Mexico. The rapid transmission of information about economic downturns to individuals and households in origin communities thus helps to explain, at least in part, why the volume of migration ceased so abruptly, and in such a synchronized fashion, with the onset of the Great Recession in the United States.

The effect of the interplay between networks and recessionary times is well illustrated by comparing the migratory behavior of households that sent out their first migrants during the period 2000–2007 (before the Great Recession) versus the period 2008–2016 (during the latter part of and after the Great Recession). Figure 3 presents a survival analysis derived from the event history of second departures from households in these two departure cohorts. As can be seen, the rates of additional departure that prevailed for households having a member in

the United States before 2008 imply that after five and a half years (66 months), one-third of all households would have sent an additional migrant northward (leaving two-thirds of the households as "survivors" that had not sent out another migrant).

In contrast, the rates of additional departure that prevailed after 2007 predict that only 22 percent of households would have sent out another member after five and a half years (leaving 78 percent of the households as "survivors"). Moreover, while the survival curve for the 2000–2007 cohort continues to decline over subsequent months, the curve of the 2008–2016 cohort remains flat with no further declines. Although other research indicates that the long-term decline and definitive end of Mexico-U.S. migration stemmed from the aging of the Mexican population (Massey, Durand, and Pren 2016), the evidence presented here suggests that the 2007 recession served as the triggering event, one that was enabled by the rapid transmission of information about the downturn to households in Mexico, which prompted them to allocate their labor effort elsewhere.

Notes

1. Taylor's proposed approach (Taylor 1986) deals with the rational behavior of a family member and the rest of his family, whereby they play a cooperative-game that produces an outcome (an agreement pay-off) that, from the point of view of any one of the players, cannot be bettered by "going his own way" because he would receive the compensation of disagreement pay-off associated with his conflict strategy. Those types of arrangement are more common among low-income households, because the economic constraints force household members to act as a unit. These households can be found in both urban and rural areas of developing countries.

2. Defined as: $f(a)+(x-a)\frac{f^{(1)}(a)}{1!}+(x-a)^2\frac{f^{(2)}(a)}{2!}+(x-a)^3\frac{f^{(3)}(a)}{3!}+\ldots$

3. This approximation is used in the portfolio based on mean-variance analysis (David 1974, 33; Eeckhoudt, Gollier, and Schlesinger 2005, 12).

4. The model of Todaro (1969) is a special case of this decision criterion, in which the individual and not the household is both the unit of decision-making and the unit of migration (Taylor 1986, 154).

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